

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1600

Burleson 138 kV
October 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1600. The Interconnection Customer proposes a 83.2 MW battery energy storage facility interconnecting at the Burleson 138 kV substation in the Pennsylvania area. The request was received on October 20, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 593 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,444,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Conemaugh Substation – Kearny 138kV	Woodland Tap – Seward 345kV L	103.6	178	9.9
Conemaugh – Canovanas Tc 138kV Line	Woodland Tap – Seward 345kV Line	101.2	186	9.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kearny – Conemaugh 138kV	Reconductor line	\$877,000	54% (\$473,000)
Conemaugh – Canovanas Tc 138kV	Reconductor line	\$1,800,000	54% (\$971,000)
	Total allocated to project		\$1,444,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.